Analysis of the Relationship Between Customer Demographics, Social Media Usage, and Marketing Campaign Performance

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As part of my Data Mining & Informatics module, I have chosen a research topic that aligns closely with both my personal and professional interests. Several years ago, I worked in social media marketing — a dynamic and continually evolving field.

Social media plays a crucial role in modern marketing strategies due to its affordability and broad reach, particularly for smaller businesses that often combine digital and traditional advertising methods.

Previously, Facebook dominated the social media landscape; however, channel preferences have diversified significantly in recent years. Based on personal observations, I hypothesised that social media usage varies across age groups: older middle-aged users (aged 40–55) may prefer Facebook, younger middle-aged users (30–40) might lean towards Instagram, while younger generations may either prefer alternative social media channels or exhibit decreased social media engagement altogether.

Since people spend a significant amount of time on social media, marketing efforts must be strategically aligned with where target audiences are most active. Understanding age dynamics is essential, as different groups engage with content differently and respond to distinct messaging styles and formats.

Given the rise of online shopping, I further hypothesised that purchasing behaviour correlates with preferred channels, where users are more likely to trust and purchase products discovered via the social networks they frequent most.

To investigate this, I selected the Cloud Enabled Marketing Strategy Dataset from Kaggle (Ziya07, 2023).

Although the dataset is synthetically generated, it realistically models real-world marketing behaviours and socio-economic patterns. In educational data mining projects — including those conducted at prestigious institutions such as Oxford and MIT — realistic synthetic data is widely accepted as "real-world" when it accurately reflects genuine behaviours and supports the application of full data mining processes.
Therefore, this dataset fully satisfies the assessment requirements.

Data Collection and Ethical Considerations

The dataset employed in this study consists of 5,000 observations across 28 variables, encompassing social media usage patterns, purchasing behaviours, and demographic characteristics such as age, gender, education level, and region. Kaggle, a globally recognised platform for research datasets, provides access to high-quality data suitable for academic analysis.

Research Objectives:

This study aims to investigate three key areas:
firstly, to explore how social media platform preferences differ across various age groups;
secondly, to examine how purchasing behaviours correlate with individuals' preferred digital communication channels;
and thirdly, to analyse how purchasing patterns and marketing campaign return on investment (ROI) vary between different age groups, with the goal of generating actionable insights for future marketing strategies.

Ethical Considerations:
The dataset is publicly available for educational purposes, contains no personally identifiable information, and complies with data protection standards. Aggregated analysis techniques were applied to prevent the identification of individuals, and efforts were made to avoid reinforcing stereotypes.

Dataset Limitations:
Although the dataset is synthetically generated, it may introduce modelling biases, as it reflects assumed real-world behaviours rather than responses from actual participants. This limitation was duly considered throughout the analysis.

1.0 Data Processing

The dataset is divided into three primary sections:

- **Demographic and personal information** (e.g., customer ID, age, gender, location, income level, education level) supporting segmentation analysis.
- **Social media activity**, including platform usage, likes, comments, shares, and time spent per week.
- **Marketing campaign performance metrics**, such as campaign ID, conversion rates, and Return on Investment (ROI).

Several variables unrelated to the research focus — such as cart abandonment rates, sentiment scores, and customer support queries — were excluded to maintain relevance and analytical clarity.

The retained data enabled analysis of demographic segmentation, correlations between social media activity and conversion rates, and the evaluation of marketing ROI across different age groups and platforms.

Following the preliminary organisation of the dataset, the next critical step involved ensuring the quality and consistency of the data through comprehensive cleaning procedures.

1.1 Data Cleaning

The dataset underwent thorough checks for missing values, with none identified, thus no imputation was necessary.
Outlier detection using descriptive statistics and the Interquartile Range (IQR) method revealed no significant anomalies.
Duplicate records were also absent.
Given that the project is exploratory rather than predictive, no further transformations (such as scaling or encoding) were required.
The dataset was therefore considered clean and ready for analysis.

2.0 Initial Data Exploration

Note on Table and Figure Numbering:
Throughout this report, the numbering of tables and figures follows the structure of the accompanying Jupyter Notebook. Sections are identified numerically (e.g., 2.1.1) to reflect different stages of the analysis. Within each subsection, multiple tables and figures may appear, distinguished by their respective titles rather than by additional numerical extensions.

Exploratory Data Analysis (EDA) was conducted to develop a deeper understanding of the dataset.
The initial exploration focused on descriptive statistics, visualisations, correlation analyses, and demographic breakdowns, providing an essential foundation for subsequent analytical steps.

2.1 Descriptive Statistics and Completeness

Descriptive statistics and completeness checks were conducted. No missing values were identified, confirming data integrity.

The average user age, considering only individuals aged 18 and above, was 41 years. Users posted 254 comments, made 149 shares, and received 499 likes

weekly, spending 25.5 hours on social media. Customers averaged 15 monthly purchases with an order value of $250.9, utilising 0.5 discounts per purchase.
Advertising metrics included $5,080 advertising spend, a 49.3% conversion rate, and a 76.9% ROI.

2.1.2 Standard Deviation Analysis

The standard deviation was 13.8 years for age, $2,843 for advertising spend, and 72.5% for ROI, indicating reasonable variability for segmentation and trend analysis.

2.1.3 Quartile Analysis

The 25th, 50th, and 75th percentiles for age were 29, 41, and 54 years.
Order values ranged from $128.8 to $372.4, and advertising spends from $2,614 to $7,525.
ROI ranged from 14.1% to 139.4%, showing significant dispersion across campaigns. These results established a foundation for segmentation.

2.1.4 Reviewing Quartile Values

Quartile values for key numerical features were assessed.
User age percentiles were 29, 41, and 54 years.
Likes ranged from 247.75 to 749, comments from 130 to 381, and shares from 76 to 223.
Order values spanned $128.81 to $372.40, with monthly purchases between 7 and 22.
Advertising spend ranged from $2,614 to $7,525, and ROI from 14.1% to 139.4%.

Overall, distributions were balanced, although ROI and ad spend exhibited greater variability.

2.2 Demographic and Behavioral Patterns Across Social Media Platforms

2.2.1 Relationship Between Age Groups and Platform Usage

The hypothesis proposed that older middle-aged individuals (40–55) primarily use Facebook, while younger middle-aged individuals (30–40) prefer Instagram, a relationship now subject to empirical investigation.

Data were grouped by age and platform, and average engagement metrics were calculated.

Table 2.2.1. Average Weekly Engagement Metrics Across Age Groups

age_group	time_spent_per_week	likes_received	comments_posted	shares
18-25	26.02	507.44	260.16	154.54
26-35	25	492.54	248.52	153.16
36-45	26.5	502.32	256.9	151.38
46-55	25.09	508.98	250.54	143.44
56-65	25.14	486.46	255.49	144.16

Data were grouped by age and platform, and average engagement metrics were calculated, as shown in Table 2.2.1. Average Weekly Engagement Metrics Across Age Groups.

Findings revealed that time spent on social media was broadly consistent across age groups, averaging 25–26 hours per week.

Minor differences were observed, with the 36–45 age group spending slightly more time and users over 46 marginally less.

Engagement metrics such as likes, shares, and comments showed minimal variation, with only a modest decline in commenting with age.

To verify whether the minimal variations in social media engagement across age groups were statistically significant, a one-way ANOVA test was conducted. The results indicated that the differences were not statistically significant ($F = 2.33$, $p = 0.054$), confirming that the observed variations were likely due to random fluctuations rather than meaningful group differences.

Overall, the analysis suggests that social media engagement intensity remains stable across generations, with older users maintaining high levels of activity.

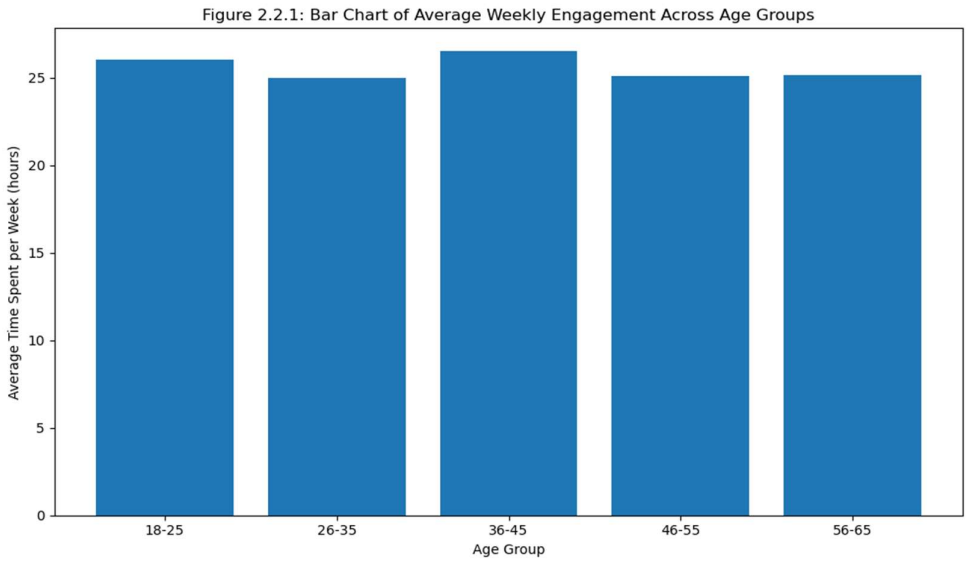

Figure 2.2.1. Bar Chart of Average Weekly Engagement Across Age Groups

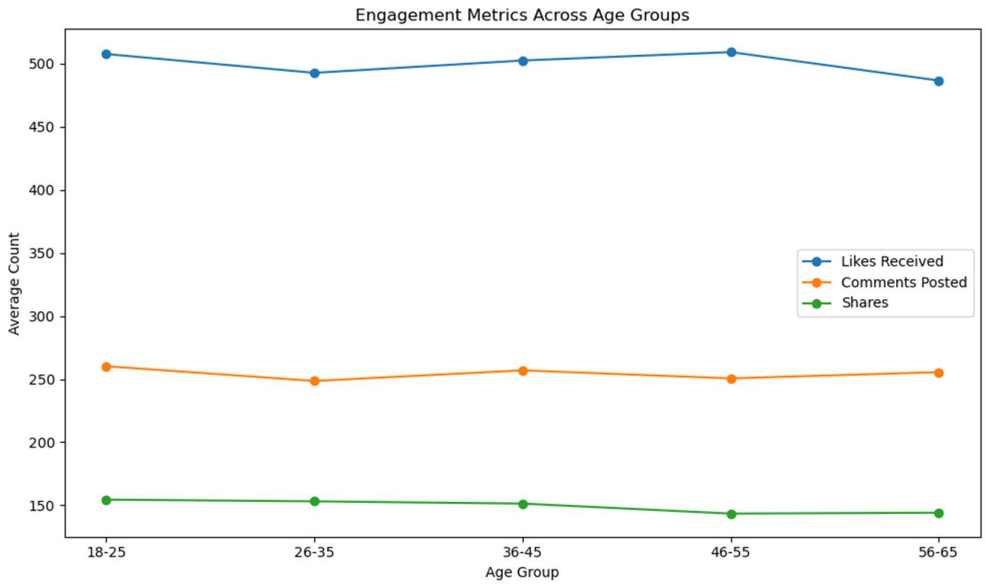

Figure 2.2.1. Line Chart of Engagement Metrics Across Age Groups

The visualisations (Figure 2.2.1. Bar Chart of Average Weekly Engagement Across Age Groups and Figure 2.2.1. Line Chart of Engagement Metrics Across Age Groups) provided a clear representation of the data presented in the tables, illustrating engagement patterns across different age groups.
Gender distribution was balanced across all cohorts (see Appendix, Table Gender Distribution Across Age Groups and Bar Chart Gender Distribution Across Age Groups – Bar Chart).
No significant disparities emerged between female, male, and "Other" gender identities.
The "Other" category encompassed non-binary individuals and those opting not to disclose their gender.

Income distribution appeared relatively uniform across age groups (see Appendix, Table Income Level Distribution Across Age Groups and Bar Chart Income Level Distribution Across Age Groups – Bar Chart), with only a slight decline in high-income users aged 56–65, potentially reflecting reduced online activity among wealthier older individuals.

Platform preferences by education level showed minimal variation (see Appendix, Table Social Media Platform Usage Across Education Levels and Bar Chart Social

Media Platform Usage Distribution Across Education Levels – Bar Chart).
LinkedIn usage was marginally lower among users with either secondary or PhD qualifications, and Instagram usage slightly higher, but differences remained minor.

Regional patterns were generally stable across age groups (see Appendix, Table Regional Distribution Across Age Groups and Bar Chart Regional Distribution Across Age Groups – Bar Chart).
Minor deviations were noted, including slightly higher engagement among Indian users aged 18–25 and 56–65, and lower engagement among 36–45 and 56–65 groups in the USA.

Finally, the initial hypothesis was reassessed.
As shown in Table 2.2.1. Platform Usage Across Age Groups – Combined Absolute and Percentage Values, platform preferences were largely uniform across age groups.
Minor shifts were observed, such as slightly elevated LinkedIn usage among 36–45-year-olds and higher Instagram usage among 18–25-year-olds; however, these differences were negligible (within 1–2%).
Thus, it was concluded that age does not significantly influence social media platform choice, and the original hypothesis was not supported.

Table: 2.2.1. Platform Usage Across Age Groups – Combined Absolute and Percentage Values

age_group	Facebook	Instagram	LinkedIn	Twitter
18-25	211 (24.0%)	242 (27.6%)	218 (24.8%)	207 (23.6%)
26-35	228 (22.1%)	269 (26.1%)	253 (24.5%)	281 (27.3%)
36-45	272 (26.0%)	258 (24.7%)	272 (26.0%)	244 (23.3%)
46-55	261 (26.0%)	252 (25.1%)	230 (22.9%)	260 (25.9%)
56-65	265 (25.4%)	280 (26.9%)	232 (22.3%)	265 (25.4%)

This similarity was further illustrated using chord diagrams to provide a clearer visual representation.

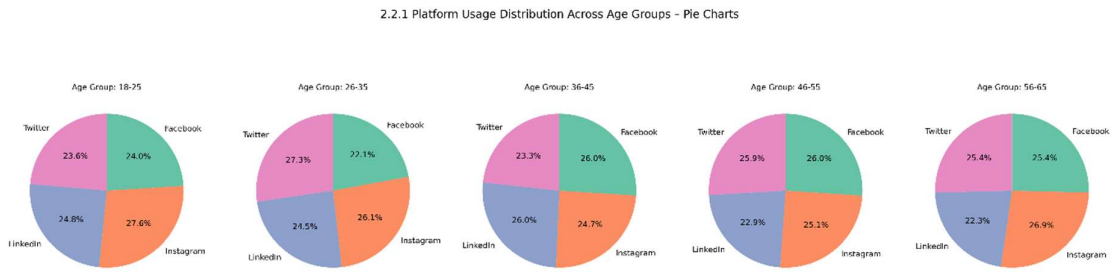

Figure: 2.2.1. Platform Usage Distribution Across Age Groups – Pie Charts

Towards the end of Section 2.2.1. regional differences in social media platform usage were assessed.

The analysis revealed minimal variations.

In Canada, LinkedIn was most preferred (26.21%), while in Germany, Instagram and Twitter were equally popular (26.47% each).

Indian users engaged slightly less with Facebook (23.20%), Instagram dominated in the UK (26.45%), and Facebook remained most popular in the USA (26.19%).

Overall, regional factors appeared to have little influence on platform choice.

These findings are summarised in Table Social Media Platform Usage by Region (Counts and Percentages) (Appendix) and visualised in Figure 2.2.1. Social Media Platform Usage by Region – Heatmap.

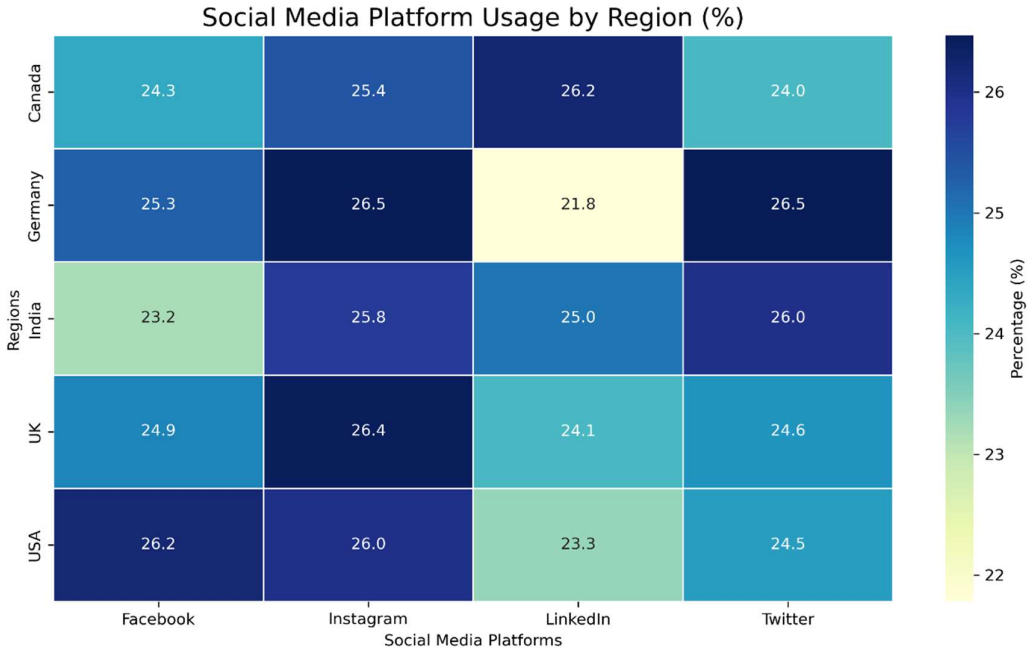

Figure 2.2.1. Social_Media_Platform_Usage_By_Region_Heatmap

2.2.1 Summary

This section examined social media usage patterns across regions, genders, age groups, and income levels.

No substantial differences were identified.

The original hypothesis — that older middle-aged individuals (40–55 years) would primarily use Facebook, younger middle-aged individuals (30–40 years) would favour Instagram, and younger generations would shift to alternative platforms or disengage — was not supported.

Across all demographics, platform usage remained largely consistent, with similar engagement patterns among both younger and older users, and no notable shift towards Facebook among older cohorts.

2.2.2 Business Perspective on Customer Behaviour

This section analysed customer behaviour from a business perspective, focusing on profitability and marketing efficiency.
Following the COVID-19 pandemic, ongoing economic disruption — including rising fuel costs, transportation expenses, and climate-related challenges — has reduced consumer spending (BBC News, n.d.).
Although wages increased modestly, they failed to match inflation, intensifying competition and elevating the importance of effective marketing.
Successful strategies now demand a thorough understanding of consumer behaviour, careful channel selection, and cost management.
Social media platforms, offering cost-effective reach, have become essential for smaller businesses.
Advertising spend, conversion rates, and ROI were analysed to assess engagement outcomes.

Initially, monthly purchasing patterns were examined by age group (see Appendix, Monthly Customer Behaviour Analysis by Age Group).
Average purchase values ranged from $239.25 to $259.95, with purchase frequency between 14.5 and 15 times per month.
The youngest group (18–25) recorded the highest average spend ($259.95), while the 26–35 group showed the lowest ($239.25).

Regional purchasing behaviour was then assessed (see Appendix, Monthly Customer Behaviour Analysis by Region).
Results revealed no substantial differences, with Germany displaying a marginally higher average purchase value.

Subsequently, purchasing behaviour was evaluated across different communication channels (see Appendix, Monthly Customer Behaviour Analysis by Platform and Region).
Distributions were largely uniform, with only minor deviations, such as slightly lower LinkedIn purchasing in Germany (21.8%), despite a marginally higher average order value ($247.90).

Overall, regional and channel-based differences were minimal, confirming that geography had little influence on purchasing patterns.
Consequently, the second hypothesis — that purchasing behaviour would align with preferred social media platforms — was not supported by the data.

These findings are further illustrated in Figure 2.2.2. Platform Usage Distribution Bubble Chart.

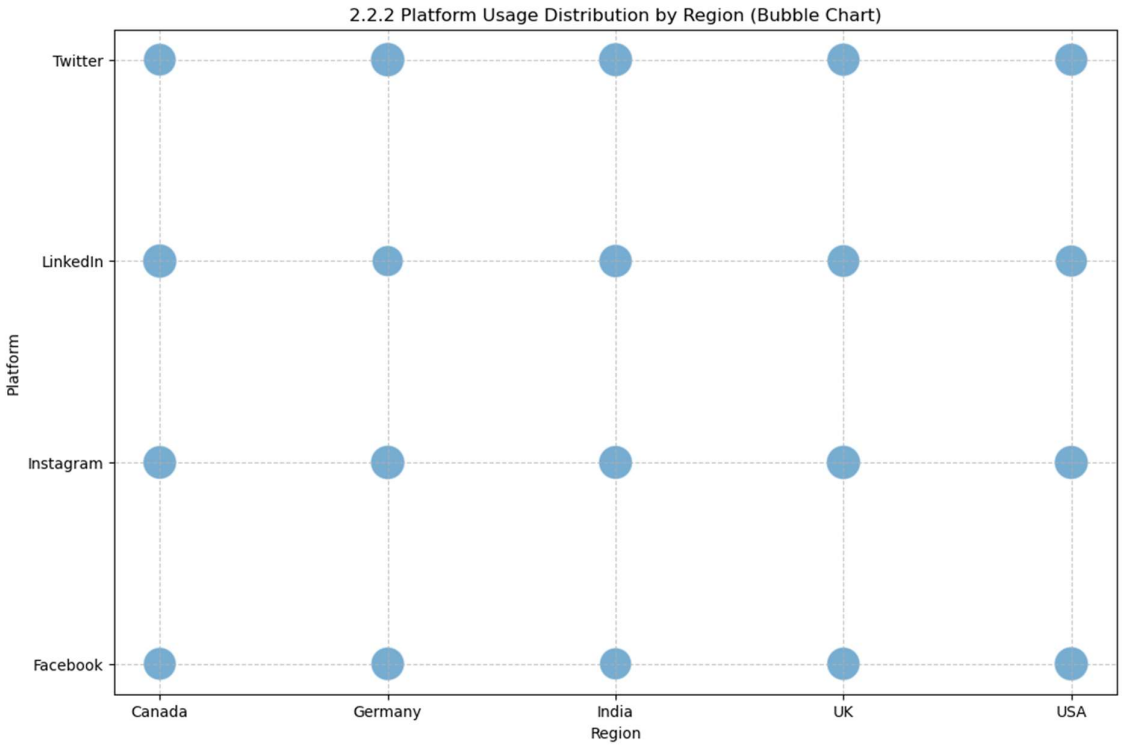

Figure 2.2.2. Platform Usage Distribution Bubble Chart

Purchase Value Distribution Analysis

At the conclusion of the customer behaviour analysis, purchase value distribution was examined to assess whether wide variation influenced the elevated mean order value.
Values ranged from $5.01 to $499.84, producing a high mean of $250.92.
To explore this further, purchases were grouped into value ranges (see Appendix,

Distribution of Average Order Value Ranges).
A substantial number of transactions occurred between $5 and $100, while most purchases were concentrated between $101 and $400.
Notably, a significant volume of high-value transactions ($401–$500) also contributed to the elevated mean.

From a marketing perspective, understanding consumer behaviour across digital media channels is critical.
Thus, the mean average order value per channel was examined (see Appendix, Mean Average Order Value by Platform).
Both the table and the line plot (Figure 2.2.2. Mean Average Order Value by Media Platform) showed relatively consistent average order values across platforms.

Consequently, the overall mean value was deemed valid for subsequent analyses, with no significant skewness detected.

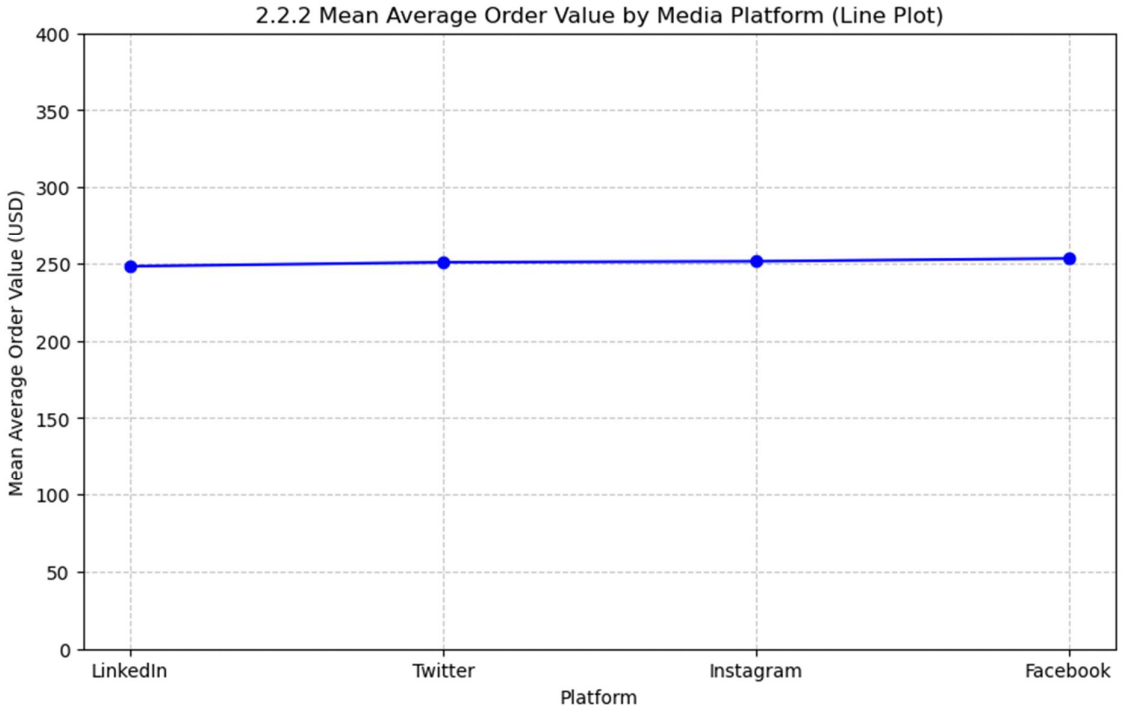

Figure 2.2.2. Mean Average Order Value by Media Platform (Line Plot)

2.2.2 Summary

This section analysed purchasing behaviour from a business perspective, focusing on spending patterns across age groups, regions, and digital channels.
Findings indicated minimal variation in purchase frequency and average order value across demographics.
Regional and channel-based purchasing behaviour also appeared relatively uniform.
Although the average order value initially seemed high, further investigation revealed that a small number of high-value purchases skewed the mean.
As the average order value did not have substantial significance for the subsequent analysis, it was accepted as a general reference point.

Section 2.2.3 Advertising Expenditure and ROI Analysis

This section explored companies' advertising expenditures, returns, and the development of Return on Investment (ROI).
The objective was to evaluate how effectively marketing spend contributes to business success—particularly crucial for small enterprises reliant on efficient investments and rapid returns to maintain competitiveness.
"Table 2.3.1: Ad Spend and ROI per Campaign (Sorted by ROI)" demonstrated significant variability in advertising costs, with campaign returns appearing random at first glance.
Subsequently, I sought underlying patterns or relationships, assuming that some logical structure must exist.
"Table 2.3.1: Ad Spend, ROI, and Platform per Campaign (Sorted by ROI)" further confirmed the absence of a clear relationship between advertising spend, marketing channel, and ROI; notably, both the lowest and highest ROI campaigns utilised TV advertising.
In an attempt to explain the randomness, I extended the analysis to examine the proportion of targeted users converting into customers.
"Table 2.3.1: Ad Spend, ROI, Marketing Channel, and Purchasers per 100 Users (Sorted by ROI)" again showed no evident correlation.

For example, Campaign 17 spent $8,758.45, achieved only a 2.59% conversion rate, and recorded a negative ROI of -48.59%.

Conversely, Campaign 10, with a modest spend of $1,598.68, achieved a 50.17% conversion rate and a positive ROI of 196.93%.

Given the lack of visible relationships, I proceeded to apply clustering techniques to search for latent structures among campaigns.

Based on the Elbow Method curve (Figure 2.3.1. Elbow Method Curve to Determine the Optimal Number of Clusters), three clusters were initially identified as optimal for segmenting marketing campaigns.

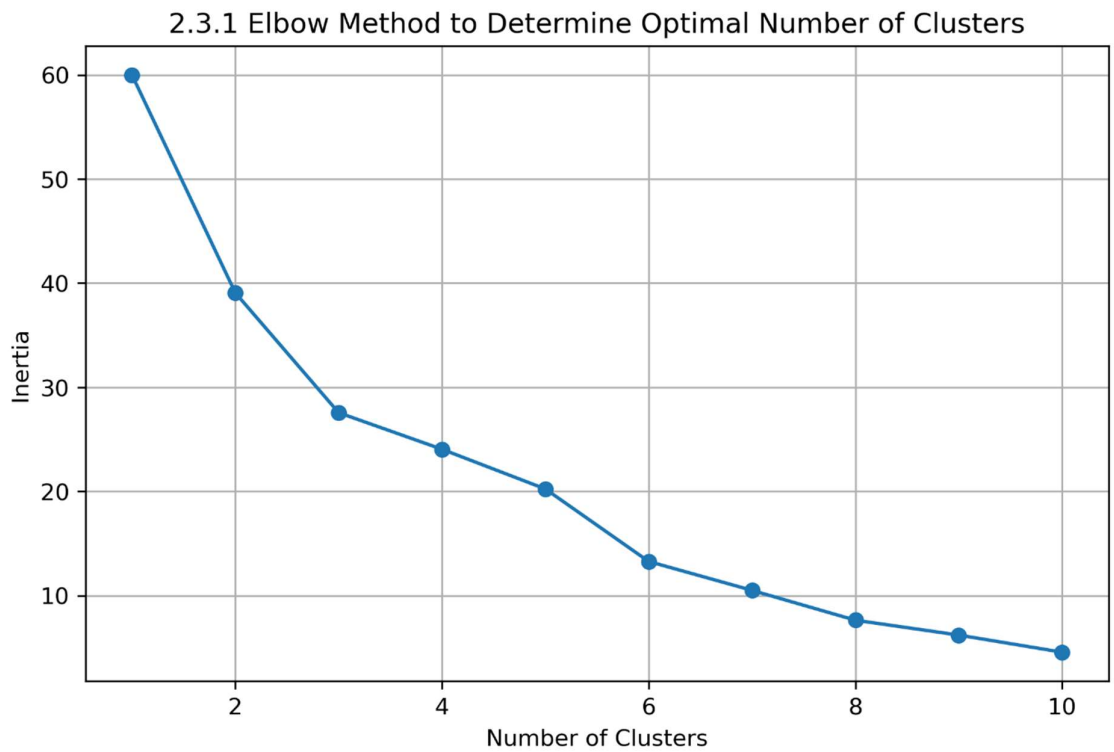

Figure 2.3.1. Elbow Method Curve to Determine the Optimal Number of Clusters

Cluster Analysis Based on K-Means

Three clusters were initially selected as the starting point for K-Means clustering. Accordingly, I applied the K-Means clustering algorithm in search of underlying patterns or correlations.

However, with three clusters, the K-Means algorithm failed to form any meaningful groupings (see Appendix: Campaign Clusters Based on Ad Spend and ROI).

In a further attempt to detect structure within the data, I adjusted the number of clusters.

Given the dataset's relatively small size, I avoided excessive fragmentation and instead tested clustering with two groups (see Appendix: Campaign Clusters with 2 Groups (Ad Spend vs ROI)).

Yet, even with two clusters, no clear segmentation emerged, prompting a final attempt with four clusters.

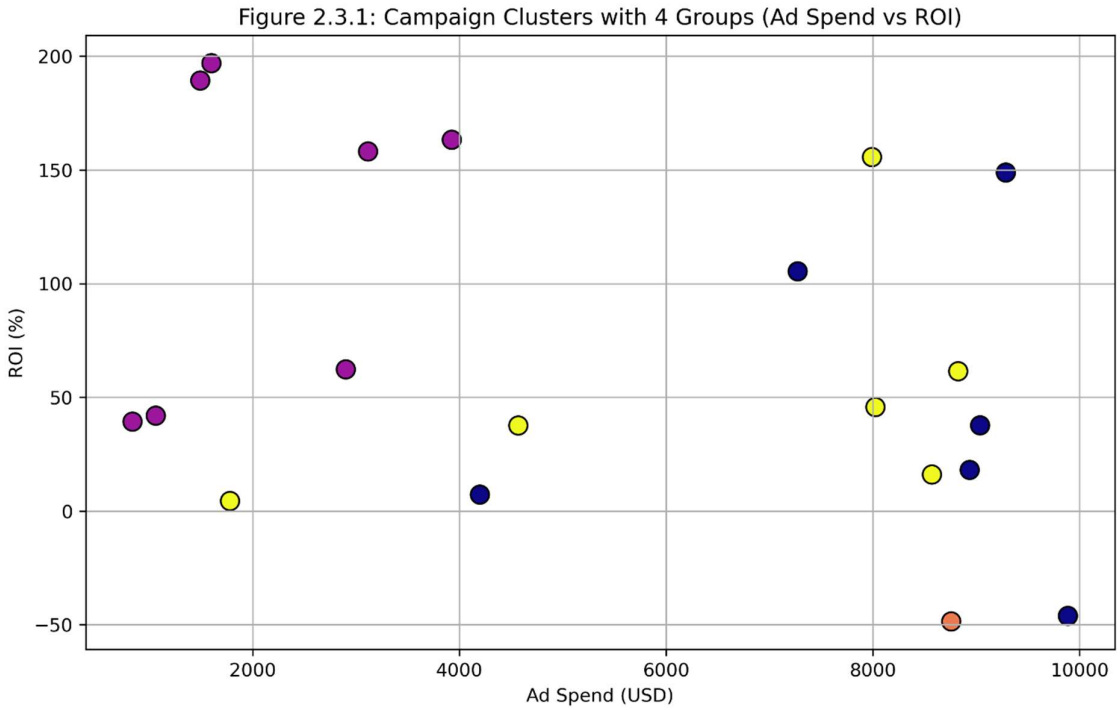

Figure 2.3.1. Campaign Clusters with 4 Groups (Ad Spend vs ROI)

The four-cluster K-Means model (Figure 2.3.1. Campaign Clusters with 4 Groups (Ad Spend vs ROI)) also failed to reveal meaningful segmentation, confirming the absence of significant relationships among variables.
Thus, marketing campaign success cannot be reliably predicted from the available financial data alone.
Key determinants likely reside in advertisement quality—particularly message effectiveness and consumer demand.

2.3 Summary

This section analysed advertising expenditure (Ad Spend), return on investment (ROI), conversion rates, and marketing channels (Social Media, Email, TV Ads, Google Ads).
No clear patterns linked channel choice, spending, and outcomes.
Cluster analysis using the Elbow Method and K-Means (with three, two, and four clusters) also failed to form distinct groupings, highlighting data dispersion.
These findings suggest that campaign success depends not only on measurable indicators but also on emotional and creative factors.
Consumer responses are often shaped by tone, visuals, and messaging—elements not captured in the dataset.

Final Summary

In this study, I explored factors influencing marketing campaign success through data mining techniques, with a particular focus on consumer behaviour. Given the intensifying competition, understanding consumer preferences has become increasingly critical to marketing effectiveness.

Two initial hypotheses were proposed: first, that platform preferences would vary by age group, and second, that purchasing behaviours would correlate with preferred digital channels.

However, detailed demographic and behavioural analyses did not substantiate either hypothesis.
Platform usage was largely consistent across age groups, and purchasing behaviours showed no strong association with specific communication channels.

Although advertising expenditure and ROI were quantifiable, the findings revealed that a substantial part of marketing success is driven by emotional factors — such as the resonance of campaign messaging and fluctuating consumer demand — which were not captured by the available data. Further research should explore these emotional and psychological dimensions; however, such granular data were not accessible within the present dataset.

Consequently, based on the current analysis, it is not possible to reliably predict purchasing patterns or marketing campaign ROI across age groups, as outcomes are significantly influenced by qualitative factors beyond standard quantitative measurement.

In conclusion, the findings reinforced that assessing marketing campaign success is inherently complex, requiring the integration of both quantitative performance indicators and qualitative, emotionally-driven consumer insights.

Appendices

Tables and Figures

- **Table 2.2.1**: Gender Distribution Across Age Groups
- **Figure 2.2.1**: Gender Distribution Across Age Groups – Bar Chart
- **Table 2.2.1**: Income Level Distribution Across Age Groups
- **Figure 2.2.1**: Income Level Distribution Across Income Groups – Bar Chart
- **Table 2.2.1**: Social Media Platform Usage Across Education Levels
- **Figure 2.2.1**: Social Media Platform Usage Distribution Across Education Levels – Bar Chart
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- **Figure 2.2.1**: Regional Distribution Across Age Groups – Bar Chart
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- **Table 2.3.1**: Ad Spend, ROI, Marketing Channel, and Purchasers per 100 Users (Sorted by ROI)
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- **Figure 2.3.1**: Campaign Clusters with 2 Groups (Ad Spend vs ROI)
- **Figure 2.3.1**: Campaign Clusters with 4 Groups (Ad Spend vs ROI)

Data Source

- Ziya07 (2023) *Cloud Enabled Marketing Strategy Dataset*. Kaggle. Available at: https://www.kaggle.com/datasets/ziya07/cloud-enabled-marketing-strategy-dataset (Accessed: 19 April 2025).